

Prior Sensitivity Analysis framework for Bayesian Marketing Mix Models (MMM)

Base Stage Project Report

Objective

The objective of the base stage is to evaluate the sensitivity of Marketing Mix Model (MMM) results to changes in a single prior assumption within a Bayesian framework. Using Google Meridian, we systematically vary one prior parameter at a time, specifically the prior mean of a selected channel's ROI, while holding all other model specifications constant. For each prior setting, the model is re-estimated and key outputs such as posterior ROI estimates and uncertainty intervals are recorded. This preliminary stage aims to quantify how strongly model conclusions depend on prior assumptions and to assess whether the data sufficiently dominates the prior or whether results remain heavily influenced by subjective beliefs.

Experimental Design

Data

We used the dataset `monthly_mocha.csv` as our dataset, this set contains monthly marketing and performance data. The dataset includes:

- Monthly revenue (our dependent variable)
- Monthly media spend by channel (e.g., Meta, Snapchat, Google, etc)
- A time index at the monthly level

Each observation represents one month. The model estimates how each channel contributes to revenue over time.

Model Setup

We used Google Meridian's model configuration for our baseline model. This model estimates the return on investment (ROI) for each media channel using Bayesian framework.

In our Bayesian model, each channel's ROI parameter has a prior distribution. The prior represents our belief about the ROI before fully learning from the data.

For example:

- We might believe Meta's ROI is around 2.0, or that Google's effects persist for several weeks.

In this base level model we implement a fixed set of prior values for each channel.

Our Prior Sensitivity Strategy

Now the goal of this project is to test how sensitive our results are to a change in such prior assumptions.

However, instead of changing the entire model at once, we:

Adjusted one channel's prior at a time and kept all other priors fixed. Then we re-ran the model and recorded how the posterior results changed

For each channel, we changed the prior mean by increments of .2 such that we tested our channels on the priors: (.2, .4, .6, .8).

What We Are Measuring

For each model run, we extracted:

- Estimated ROI's at each prior mean input

Now using this information we can make much more formal conclusions about the prior sensitivity of our model.

Purpose of This Experiment

Using this setup we are able to answer questions like:

- If our prior belief is wrong by 10%, 25%, or even 50%, do our conclusions change?
- Are some channels more sensitive to prior assumptions than others?
- Does the data dominate the prior, or does the prior meaningfully influence results?

✓ **Summary of the Model Output**

Below is the output which we obtained from our model:

target_channel	roi_prior_mu	channel	estimated_roi
meta	0.2	meta	0.09353103489
meta	0.2	google	0.05846856162
meta	0.2	snapchat	0.1046568528
meta	0.4	meta	0.104739964
meta	0.4	google	0.06685409695
meta	0.4	snapchat	0.08943032473
meta	0.6	meta	0.09881845862
meta	0.6	google	0.05909004062
meta	0.6	snapchat	0.09728041291
meta	0.8	meta	0.1048809513
meta	0.8	google	0.05197326466
meta	0.8	snapchat	0.1052171141
google	0.2	meta	0.0898340717
google	0.2	google	0.06196143478
google	0.2	snapchat	0.0949396044
google	0.4	meta	0.104739964
google	0.4	google	0.06685409695
google	0.4	snapchat	0.08943032473
google	0.6	meta	0.09372347593
google	0.6	google	0.06711129844
google	0.6	snapchat	0.09271331877
google	0.8	meta	0.09774143994
google	0.8	google	0.06474275142
google	0.8	snapchat	0.09839548916
snapchat	0.2	meta	0.1099815518
snapchat	0.2	google	0.06829557568
snapchat	0.2	snapchat	0.1080736965
snapchat	0.4	meta	0.104739964
snapchat	0.4	google	0.06685409695
snapchat	0.4	snapchat	0.08943032473
snapchat	0.6	meta	0.09385755658
snapchat	0.6	google	0.05814940855
snapchat	0.6	snapchat	0.106964983
snapchat	0.8	meta	0.09590141475
snapchat	0.8	google	0.05872449279
snapchat	0.8	snapchat	0.1092617065

```
from google.colab import files
uploaded = files.upload()
```

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Saving prior sensitivity results csv to prior sensitivity results (2) csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Loading our CSV
results = pd.read_csv("prior_sensitivity_results.csv")
```

```

target_only = results[results["channel"] == results["target_channel"]]

g = sns.FacetGrid(
    target_only,
    col = "target_channel",
    col_wrap = 2,
    height = 4,
    aspect = 1.2,
    sharey = False
)

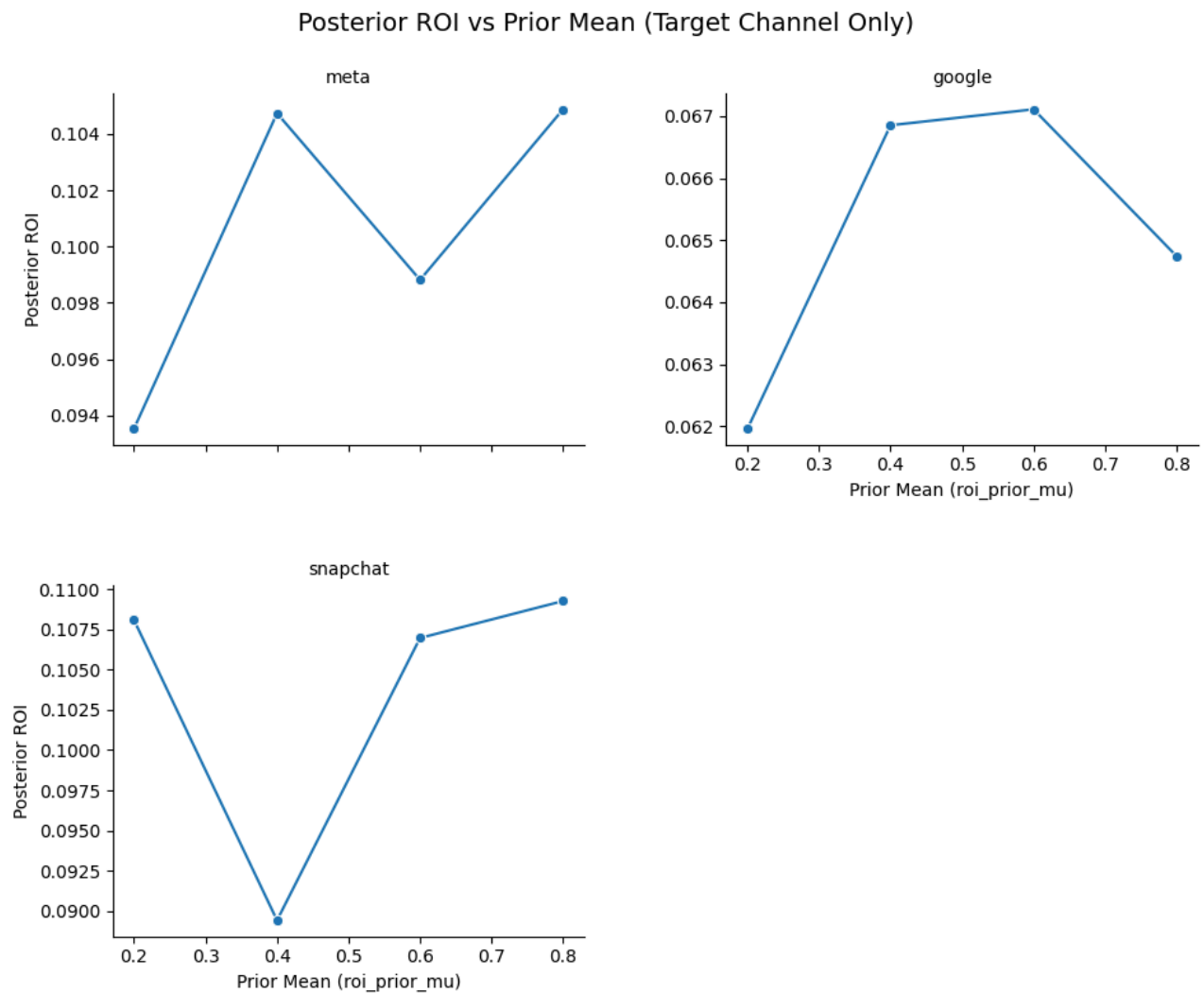
g.map_dataframe(sns.lineplot, x="roi_prior_mu", y="estimated_roi", marker="o")

g.set_axis_labels("Prior Mean (roi_prior_mu)", "Posterior ROI")
g.set_titles("{col_name}")

plt.subplots_adjust(top=0.9, hspace=0.4, wspace=0.3)
g.fig.suptitle("Posterior ROI vs Prior Mean (Target Channel Only)", fontsize=14)

plt.show()

```



```

from sklearn.linear_model import LinearRegression

slopes = []

for (tgt, ch), group in results.groupby(["target_channel", "channel"]):
    X = group[["roi_prior_mu"]].values
    y = group["estimated_roi"].values

    model = LinearRegression().fit(X, y)
    slope = model.coef_[0]

    slopes.append({
        "target_channel": tgt,
        "channel": ch,
        "slope": slope
    })

sens_slopes = pd.DataFrame(slopes)
sens_slopes = sens_slopes.sort_values(by="slope", key=abs, ascending=False)

sens_slopes

```

	target_channel	channel	slope
7	snapchat	meta	-0.026561
6	snapchat	google	-0.018709
4	meta	meta	0.014064
3	meta	google	-0.013625
8	snapchat	snapchat	0.010549
2	google	snapchat	0.006825
1	google	meta	0.006353
5	meta	snapchat	0.004765
0	google	google	0.004301

The table above allows us to see the correlation (as a slope) between how the prior mean scales the posterior mean. However, these values are not too impactful, i.e. all slopes are near to zero. Meaning this information is rather unimpactful likely as a results of channel spill over.

```

range_stats = []

for (tgt, ch), group in results.groupby(["target_channel", "channel"]):
    group = group.sort_values("roi_prior_mu")

    post_change = group["estimated_roi"].iloc[-1] - group["estimated_roi"].iloc[0]

    range_stats.append({
        "target_channel": tgt,
        "channel": ch,
        "posterior_change": post_change
    })

sens_range = pd.DataFrame(range_stats)
sens_range = sens_range.sort_values(by="posterior_change", key=abs, ascending=False)

```

sens_range

	target_channel	channel	posterior_change
7	snapchat	meta	-0.014080
4	meta	meta	0.011350
6	snapchat	google	-0.009571
1	google	meta	0.007907
3	meta	google	-0.006495
2	google	snapchat	0.003456
0	google	google	0.002781
8	snapchat	snapchat	0.001188
5	meta	snapchat	0.000560

So we can now see the difference that the prior has on the channel and the change that the posterior has as a result.

Key Findings

We varied the ROI prior mean from *0.2* to *0.8* for each channel.

Meta ROI Prior Sensitivity

The posterior Meta ROI estimated ranged from *0.0935* to *0.1049*, reflecting a **12.13% increase** across the tested prior range. The movement is smaller than the magnitude of the prior shift itself, indicating moderate prior sensitivity.

Some cross-channel effects were also observed. Google ROI under Meta prior variation increased from *0.0520* to *0.0669*, while Snapchat ROI increased from *0.0894* to *0.1052*

Google ROI Prior Sensitivity

The posterior Meta ROI estimated ranged from *0.0620* to *0.0671*, corresponding to a **8.31% increase** across the tested prior range. The relative change is less than the relative change of Meta, indicating that Google channel is more robust to prior assumptions within the tested range.

Other channels showed fluctuation during this process. Meta ROI ranged from *0.0898* to *0.1047* and Snapchat ROI ranged from *0.0894* to *0.0984*.

Snapchat ROI Prior Sensitivity

The posterior Snapchat ROI estimated ranged from *0.0894* to *0.1093*, which is a **22.18% increase** across the tested prior range. The relative change is nearly double the sensitivity observed for Meta and almost three times that of Google, suggesting a strong prior sensitivity.

In addition, Meta's ROI shifted between *0.0939* and *0.1100* under Snapchat prior variation.

Interpretation

Based on percent change in posterior ROI under equivalent prior shifts, Snapchat is the most prior-sensitive channel in the current model configuration, while Google is comparatively robust.

Across all channels, posterior ROI estimates did not shift one-for-one with prior changes, indicating that the data meaningfully constrains estimates. However, the degree of sensitivity varies by channel, and in particular Snapchat demonstrates substantial dependence on prior specification.

Practical Implications

- **Interpretation of Prior Influence**

The sensitivity analysis shows that posterior Meta ROI changes by approximately 12.1% when the prior mean varies from 0.2 to 0.8. This indicates that the prior influences the posterior estimate, but does not dominate it. The observed data contributes meaningful information, and the likelihood partially updates prior beliefs.

This suggests that the model demonstrates reasonable posterior stability within the tested prior range.

- **Implications for Budget Allocation**

Marketing investment decisions rely on accurate ROI estimation. If posterior ROI estimates are highly sensitive to prior assumptions, budget allocation recommendations may lack robustness.

By quantifying how ROI responds to changes in prior specification, this framework provides a measure of decision stability. A relatively moderate sensitivity implies that strategic recommendations based on the model are less likely to change dramatically under alternative but plausible prior beliefs.

- **Model Transparency and Governance**

Bayesian models require explicit prior assumptions. Without systematic evaluation, the impact of these assumptions may remain unclear.

This framework improves transparency by explicitly linking prior parameter variation to posterior ROI outcomes. Stakeholders can observe how belief changes propagate through the model, increasing interpretability and credibility in applied marketing settings.

- **Extension to Broader Robustness Evaluation**

Although demonstrated for a single channel, the same procedure can be applied to other media channels and hyperparameters. Incorporating prior sensitivity analysis as a standard validation step can strengthen the reliability of Bayesian MMM implementations before deployment.

Limitations

Despite its methodological contributions, several limitations should be acknowledged.

- **One Dimensional Prior Perturbation**

The analysis varies only one prior parameter (Meta ROI prior mean) while holding all other priors fixed. In practice, priors may interact across channels through shrinkage or correlation structures.

A more comprehensive analysis would involve joint variation of multiple hyperparameters to capture potential interaction effects.

- **Limited Distributional Sensitivity Metrics**

The evaluation primarily examines changes in posterior mean ROI. However, full posterior distributions may also change in variance or tail behavior.